

Variational Low-Rank Adaptation Using IVON

Small-scale adaptation: Qwen2.5-0.5B on ARC-Easy
Siddhu, Pathlavath Shiva Kumar, Gugulothu Ganesh

Run completed: 14 September 2026 | Google Colab, Tesla T4
Six completed training runs | Three paired seeds | 12 correctness tests passed

Research question and method

Does uncertainty-aware LoRA training improve confidence quality in a small language model? We froze the base model and trained 540,672 rank-8 adapter parameters. AdamW learns a single adapter. IVON estimates a diagonal Gaussian distribution over adapter parameters. We evaluate IVON at its mean and by averaging probabilities from 10 posterior draws.

Our measured test results

Prediction method	Accuracy (%)	ECE (points)	NLL
AdamW-LoRA	73.18 ± 1.76	6.89 ± 2.30	0.680 ± 0.005
IVON at mean	72.14 ± 0.23	6.99 ± 0.29	0.680 ± 0.003
IVON, 10 draws	69.40 ± 1.37	17.95 ± 1.49	0.884 ± 0.040

Mean ± sample standard deviation across seeds 21, 42 and 87. The same 256 held-out test questions were used for every run. Higher accuracy is better; lower ECE and NLL are better. ECE uses 15 equal-width bins. NLL uses natural logarithms.

What the evidence supports

We did not find a clear calibration benefit for IVON mean under these settings. Its paired ECE difference from AdamW was +0.105 ± 2.394 points, and its NLL difference was -0.000165 ± 0.002448. These gaps are small relative to the observed seed variation. Ten-draw prediction had worse average accuracy, ECE and NLL.

This is a valid small-scale experimental finding. It is not an exact reproduction of the paper's six-task Llama-2 7B study and cannot isolate the effect of model size. The published results are not substituted for our measurements.

Training evidence

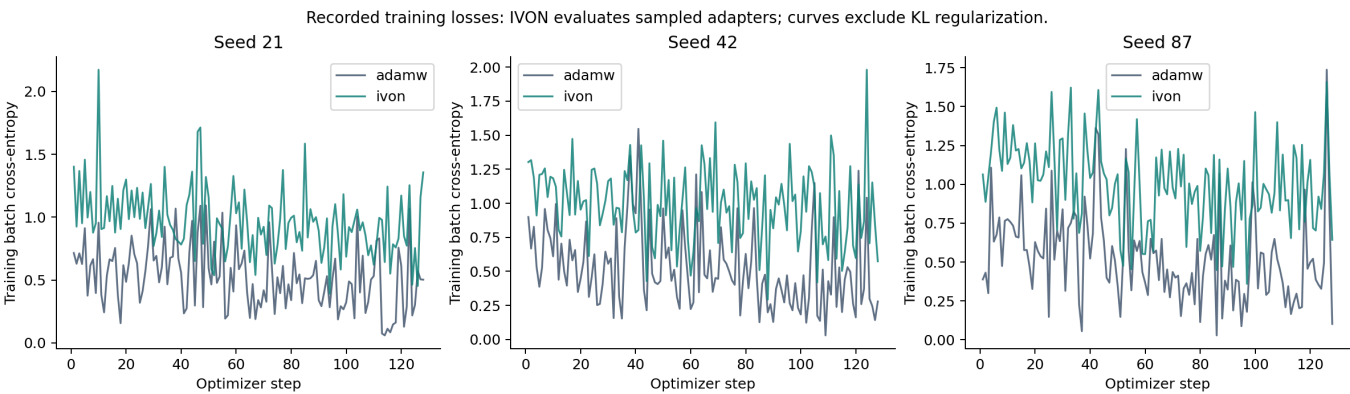
Every optimizer/seed pair completed 128 updates. The main training and evaluation pipeline took 720.8 seconds, approximately 12 minutes. Training time below excludes validation and test prediction.

Fixed protocol

Data: 512 train / 128 validation / 256 test; subset seed 2026.
Training: FP32, microbatch 4, accumulation 2, effective batch 8.
Adapters: q_proj and v_proj; rank 8; alpha 16; dropout 0.
Optimizer settings: AdamW learning rate 0.0002; IVON learning rate 0.03; both weight decay 0.0001. IVON ESS 1,000,000; Hessian initialization and clipping radius 0.001. Settings were fixed before test evaluation.

Seed	Optimizer	Completed steps	Training time	Peak allocation
21	ADAMW	128	44.3 s	2.96 GiB
21	IVON	128	50.4 s	2.96 GiB
42	ADAMW	128	51.7 s	2.96 GiB
42	IVON	128	52.6 s	2.96 GiB
87	ADAMW	128	51.9 s	2.96 GiB
87	IVON	128	52.1 s	2.96 GiB

Recorded training curves

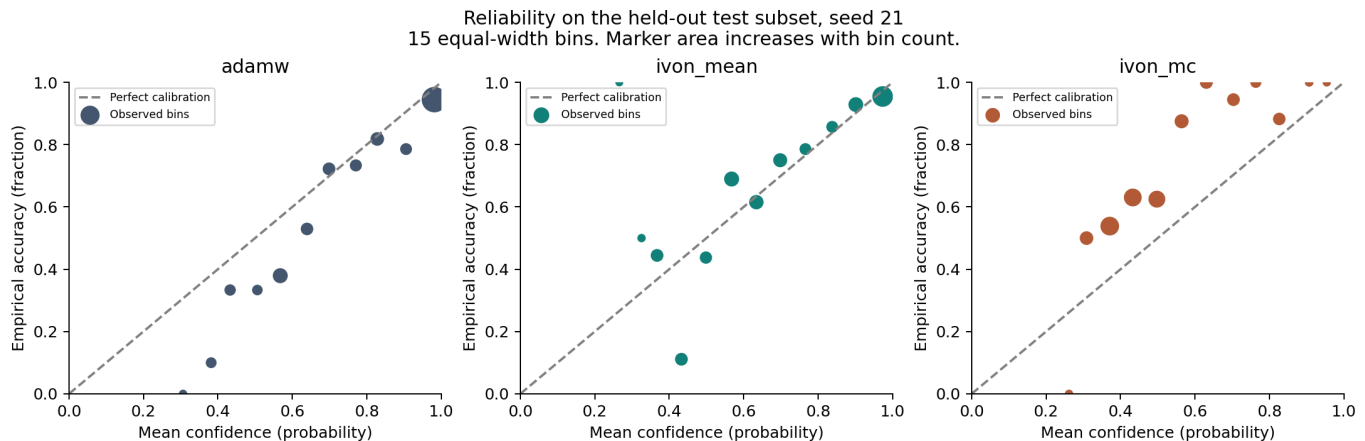


These are recorded minibatch cross-entropy losses. IVON evaluates a sampled adapter; the plotted losses exclude KL regularization. The curves therefore do not represent the same complete optimization objective. Peak memory is PyTorch CUDA allocation, not total device usage.

Evidence checks: all six histories contain 128 steps; initial adapter hashes match within paired seeds; test IDs and labels match across methods; all saved test metrics were independently recomputed from individual probabilities; packaged training source matches the source hash recorded in Colab.

Explain and demonstrate

Calibration behavior



Reliability diagrams show seed 21 only, with 15 equal-width bins. Points on the diagonal represent perfect calibration; marker size increases with bin count. This single-seed illustration does not replace the three-seed results table.

Why did sampling not help here? Across the three seeds, its average top-choice confidence was 52.1%, compared with 69.4% accuracy. The sampled predictor was underconfident overall. Lower confidence alone does not guarantee calibration. The experiment does not establish which hyperparameter or approximation caused this behavior.

Five-minute presentation order

1. Explain LoRA and the difference between accuracy and confidence.
2. Explain IVON's Gaussian adapter approximation and probability averaging.
3. Show the executed Colab notebook: Tesla T4, passing tests, six training logs.
4. Show this results table, the seed variation and recorded loss curves.
5. State the limited conclusion and propose equal-budget validation tuning as a follow-up.

Answers to prepare

Is the whole model Bayesian? No. Only the adapters have an approximate distribution.

Why report ECE and NLL? ECE measures binned confidence gaps; NLL penalizes low probability on the correct answer. Both complement accuracy.

Why three seeds? To show initialization/order variation, without claiming statistical significance.

Is uncertainty free? Variance estimation reuses IVON state, but 10-draw prediction requires ten forward passes.

What are the limits? One small task subset, three seeds, fixed initial settings, a different model family, and a cold diagonal posterior.

Sources and local evidence

Cong et al. (2024), Variational Low-Rank Adaptation Using IVON, arXiv:2411.04421.

Authors' IVON-LoRA code; official IVON optimizer.

Measured evidence: results/quick/summary/summary.json, saved per-example predictions, training histories, environment.json and local_verification.json. The project ZIP includes the executed notebook and editable presentation.